

manufacturers like Arduino, Sparkfun, Adafruit, Seedstudio and Mikroelektronika are popular among students, hobbyists and professionals for the quick prototyping tools they offer."

Shah adds, "Development boards for MCUs based on AVR, PIC, 8051 and ARM7 architectures are very popular among students and beginners as these are easily available in the local market. These come with a varying set of onboard peripherals to suit a wide range of applications for every budget."

These can be used for rapid prototyping and smaller proof-of-concept (POC) development. "Boards such as Arduino, MangoPi, chipKIT, mbed, MSP430 Launchpads and STM32 discovery kits, to name a few, fall under this category," notes Syed Anwaarullah, founder, PotentialLabs.

He adds, "MCU development platforms such as Arduino come with a host of free and open source libraries that help in building rapid prototypes very easily with the help of add-on shields and expansion boards."

Application-specific development boards. Anwaarullah informs that, such boards are for dedicated tasks mostly used by companies or engineers building specific products in areas such as home/industrial automation, automotive applications and more. He says, "These development kits allow to rapidly test the functionality of the MCU as per requirements before moving on to fabricating custom printed circuit boards (PCBs)." Such boards are usually costly in comparison to generic development boards, but offer more features and a dedicated development tool chain.

Platforms like Raspberry Pi (or Raspi), BeagleBone and mbed have very high processing power running into GHz and random access memory (RAM) that extends in GBs, remarks Neha Mehta, founder, Hacktronics India. She says, "These boards run custom Linux variants and encapsulate all complexities of the external hardware into friendly programming languages like Python, for instance."

Why is Arduino a hot favourite in entry-level MCU boards

"Popularity of an MCU development kit is primarily driven by the ease of the development environment. Arduino revolution has shown the way it has to be and has inspired many to take up embedded prototyping."

—**Darshan Virupaksha, systems engineer, Allux Innovations, and co-organiser, IoTBLR**

"It is because of the ease of programming Arduino that gets even non-programmers such as design students and hobbyists to tinker with MCUs without knowing advanced MCU C/C++ or assembly languages."

—**Syed Anwaarullah, founder, PotentialLabs**

"Arduino and its variants are perfect for beginning with programming embedded electronics. Used mostly by students and in open source projects, these are available with almost every major retailer in India. Benefits of these boards are cost, availability and libraries support. Almost all major sensors have libraries written for Arduino."

—**Neha Mehta, founder, Hacktronics India**

Industry-ready development boards.

Advanced processing power boards based on ARM7 or ARM9 architecture are used for industrial applications. "These include Cubietruck boards and dual-core or quad-core processors-powered Allwinner boards, which are mostly used for media processing in applications like smart televisions, media centres and digital video recorders (DVRs)."

Such boards can be used directly in real-time applications. Anwaarullah says, "For instance, Arduino Micro or Pro Mini are compact enough to be embedded inside products directly and can also be programmed on-the-go as requirements change."

Features in the spotlight

Different MCUs have varying features and the entire development board is usually based on the supported features of the MCU. Thushar V.P., technical director, Rhydo Technologies (P) Ltd, says, "Nowadays, many development boards such as Raspi or BeagleBone support an onboard operating system (OS). Though a great feature, these boards tend to be more expensive than the ones without an onboard OS."

Primarily, one does look at MCU boards for their processing and memory capabilities, but nowadays most of these offer decent RAM and flash capabilities, remarks Virupaksha. He says, "This makes interface [like serial peripheral interface (SPI) and Inter-IC (I2C)] and out-of-box

support offered more valuable." He adds, "Many chips now also support a universal serial bus (USB) natively and sometimes USB host as well."

Features such as availability of embedded wireless modules, dedicated functionality such as encryption and data logging, over-the-air (OTA) programming and mobile/cloud synchronisation are some new features to look out for, observes Anwaarullah.

ARM, with its mbed initiative, has helped programmers to have a variety of mbed-enabled boards using MCU chips from most of the ARM licencees like NXP, STMicroelectronics and Freescale, among others, notes Virupaksha. "A rich variety of features in terms of input/output (I/O) peripherals, low-power modes and hardware accelerator for digital signal processing (DSP) plus security-coupled software support make the development process a joyful ride with the latest development boards."

Another addition to this exciting lot is Java/Java script-enabled boards. Virupaksha adds, "Knowledge of Embedded C is no more a limitation. Traditional Java programmers can now code on MCUs. Java source code is compiled using cross compilers and programmed on to the target device." Other higher level languages like .net are supported too. Espruino development kit can be programmed using Java scripts and Netduino using .net.

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